

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Secnd Semester 2025-2026

Mid-Semester Test
(EC-2 Regular)

Course No. : AIML ZC418
Course Title : Introduction to Statistical Methods
Nature of Exam : Closed Book
Weightage : 30%
Duration : 2 Hours
Date of Exam : 14/03/2026

No. of Pages	= 2
No. of Questions	= 6

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q1. The daily income (in ₹) of 12 small shop owners is: 200, 220, 240, 260, 280, 300, 320, 350, 400, 600, 900, 1500.

(a) Calculate Q1, Q3, and the Inter Quartile range. [1.5M]

(b) Determine whether the distribution is right-skewed or left-skewed using mean and median. [1.5M]

Q2. A company is launching a new product and wants to estimate the probability of a customer buying it based on their age and income level. A sample of 1000 potential customers is collected. The age distribution is as follows: 300 customers are between 18–25 years, 400 customers are between 26–35 years, 300 customers are above 35 years.

The income distribution is: 200 customers have low income ($< ₹30,000$), 500 customers have medium income ($₹30,000–₹60,000$), 300 customers have high income ($> ₹60,000$).

Assume that age group and income are independent and each customer belongs to exactly one age group and one income group. The conditional probabilities $P(B | A_i \cap I_j)$ for all age–income combinations are given in the following table:

Age group Income	18 - 25	26 – 35	Above 35
Low	0.2	0.25	0.3
Medium	0.25	0.3	0.35
High	0.3	0.35	0.4

(a) Using the Law of Total Probability, compute the overall probability that a randomly selected customer buys the product.[3M]

(b) If a randomly selected customer has bought the product, what is the probability that the customer belongs to the age group above 35 and has high income? [2M]

Q3. Consider the following training dataset used for binary classification. Each feature F_1, F_2, F_3 can take one of three nominal values: a, b, c.

F1	F2	F3	Category
a	c	a	+
c	a	c	+
a	a	c	-
b	c	a	-
c	c	b	-

(a) Using the Naïve Bayes classifier with Laplace smoothing with smoothing constant = 1, compute:

1. The prior probabilities of each class. [1M]
2. The conditional probabilities required for classification. [3M]

(b) Using the probabilities computed above, determine the posterior probability for each class and classify the following test example:

$$F_1 = a, F_2 = c, F_3 = b$$

Clearly show all steps of your calculations and state the final predicted class. [3]

Q4. Let X be a continuous random variable with probability density function:

$$f(x) = \begin{cases} 6x^5, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Verify that $f(x)$ is a valid probability density function. [1M]
- (b) Find the mean $\mu = E[X]$. [2M]
- (c) Find the standard deviation of X . [2M]
- (d) Find the probability $P\left(\frac{1}{2} < X < \frac{3}{2}\right)$. [2M]

Q5. Let X and Y be two discrete random variables with joint probability mass function given below:

X \ Y	0	1	2
0	0.1	0.05	0.05
1	0.15	0.2	0.1
2	0.1	0.15	0.1

- (a) Find the marginal probability distributions of X and Y . [1M]
- (b) Compute: $P(X \leq 1, Y \geq 1)$ [1M]
- (d) Determine whether X and Y are independent random variables. Justify your answer. [2M]
- (e) Find $E[X]$, $E[Y]$, and $E[XY]$. [1M]

Q6. A company produces light bulbs with a probability of defect, $p = 0.05$. If 15 bulbs are selected randomly, calculate the probability of getting exactly 2 defective bulbs using the binomial distribution formula. Then, assuming n is large and p is small, use the Poisson distribution as an approximation to find the probability of getting exactly 2 defective bulbs, where $\lambda = np$. Compare the results from both distributions. [3M]
